Project - Molecular dynamics simulation: **Liquid water solvating a single carbon nanotube**

Objectives: Upload a water-carbon nanotube interface, run a full MD simulation and analyze microscopical structure and dynamics of water.

Tools: VMD (data visualization and analysis), NAMD (MD simulation)

STEP 1: Preparing Setup. Install and open VMD according the operating system of your computer —> http://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=VMD

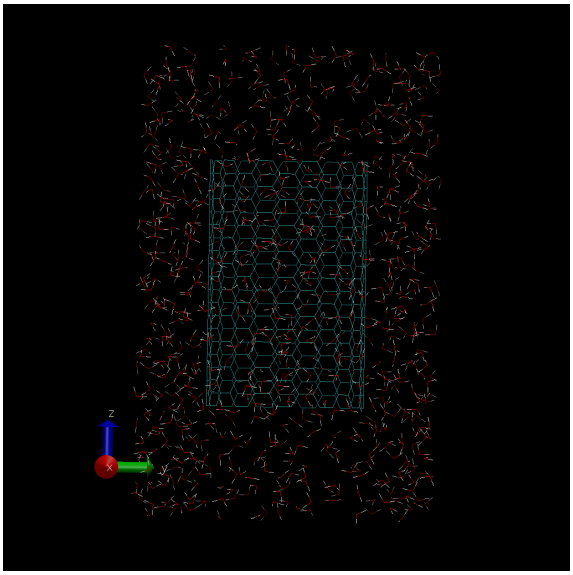
NOTE: The first time you'll need to login as a new user into the page:
http://www.ks.uiuc.edu/Research/vmd/

1. Open VMD and upload your initial setup. The hydrophobic surface will be a single c*arbon nanotube* (CNT) of the class *(n,m)*: you'll need to choose chiral indexes n and m and length L of the tube. For instance, a (11,11) tube of length 2 nm.

2. You should upload files cnt11.psf and cnt11.pdb in this order:

'File' -> 'New Molecule' -> Filename = cnt11-water.psf -> Load
'File' -> 'New Molecule' -> Filename = cnt11-water.pdb -> Load

This will show a CN (11,11) in your screen as this:



STEP 2: Equilibration. Install and open NAMD according the operating system of your computer —> http://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=NAMD

Make sure it works by trying: > namd3 or ./namd3

It will require files 'input.namd' and 'par_all27_prot_lipid.prm' to be in the working directory, where all specifications of the simulation are included. Create a RUN directory in the current folder.

Start with ***energy minimization***. Edit the 'input.namd' file as follows:

a. Comment out temperature lines (15, 20, 24)
b. Set 'coordinates' (line 38) as file "water-cnt11.pdb"
c. Comment out 'coordinates' (line 39) as directory "./RUN/FINAL.coor"
d. Switch from "off" to "on" the 'minimization' line (141)
e. Set sizes of main simulation box ('cellBasisVector1-3', lines 186-188) as:

28.0 0.0 0.0
0.0 28.0 0.0
0.0 0.0 40.0

f. Run > 'namd3 input.namd'. [Optional, strongly recommended: > 'namd3 input.namd > output.log' &' if you want to store thermodynamical data in file 'output.log']. HINT: sometimes, in unix environments, you should use ‘./namd3 …’

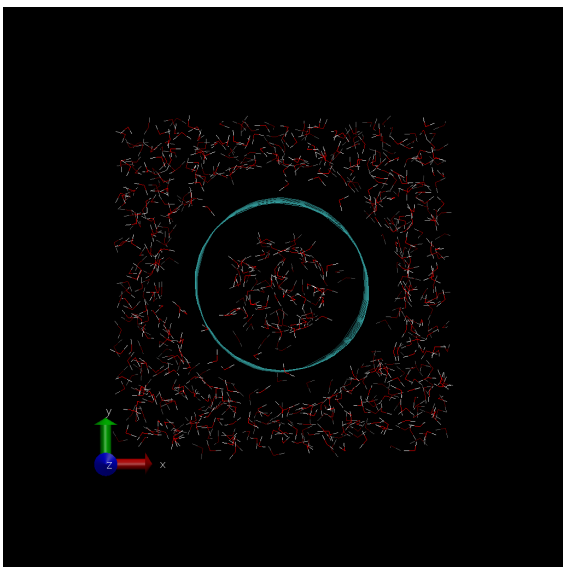
After energies have converged (between 3000-4000 steps, check output to make sure energies are roughly constant), we should run a ***MD simulation for equilibration***. To do so, we should edit again 'input.namd' as follows:

a. Activate temperature lines (15, 20, 24), use 298 K, freq.=50
b. Comment out 'coordinates' (line 38) as file “water-cnt11.pdb”
c. Set up 'coordinates' (line 39) as directory “./RUN/FINAL.coor”
d. Switch from “on” to “off” the 'minimization' line (141)
e. Keep box sizes
f. Run > 'namd3 input.namd > output.log &' for a few steps, maybe with a small timestep (0.1 fs, to gradually increase to 1 fs as the target value), but you can try 1 fs as first choice and cross fingers or, if you are religious, pray. If it works, you can try 1000 or even 10000 time steps.

IMPORTANT (and aesthetic) HINT: AFTER the simulation ends, you can visualize the trajectory of your system ('movie') using VMD: Run > vmd water-cnt11.psf ./RUN/trajectory.dcd

The system will be equilibrated when temperature, energy and pressure keep fluctuating around given values. To check it, use 'Extensions' -> 'Analysis' -> 'NAMD Energy'. Set 'Selection 1' and 'Energy type ' to 'all'.

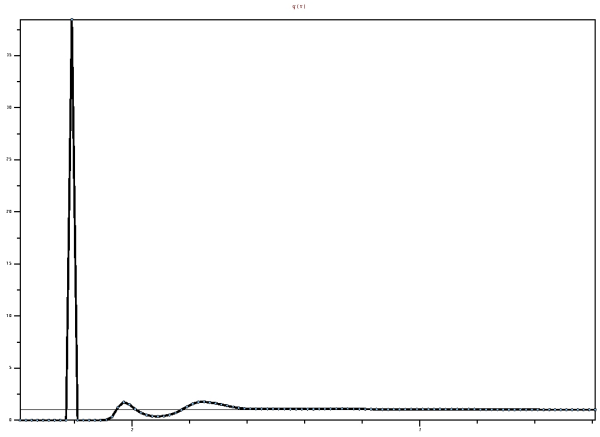
Hint: In a complementary way, you can use the tool 'NAMD Plot', which requires the previous generation of the log file during the simulation. There you can easily get energy, temperature and pressure as function of simulation time. This is the aspect of your system after equilibration:



STEP 3: MD Production run and calculation of physical properties. After the system is equilibrated, a long run is in order. Use for instance 100000 steps. Get a (big) refreshment!

After the long run ends, check the trajectory, energy, temperature and pressure again. If all of them are OK, compute the following properties:

a. **Radial distribution functions of O-O, O-H, H-H, C-O, C-H:** Use 'Extensions' -> 'Analysis' -> 'Radial Pair Distribution Function g(r)'. In 'Selection 1' and '2', write: type 'OT' for g(r)_{OO}. The numbers of frames run from 0 to N-1 (according the count shown at the VMD main screen as 'Frames'). For hydrogens use type 'HT' and type 'CA' for carbons. As an example, here is the rough g_{OH}(r):



b. **Coordination numbers**: The n(r) corresponding to the above g(r) may be obtained with the same tool ('Radial...') using 'Display Int(g(r))'.

Describe your results in physical terms.

c. **Hydrogen bonds in water:** Using the tool Extensions' -> 'Analysis' -> 'Hydrogen Bonds', calculate the number of HB per water molecule in the system. Set 'Selection 1' as 'resname TIP3' and set HB definition to 'Donor-Acceptor distance = 3.6 Å' and 'Angle cutoff = 30'.

Knowing that bulk water has about 4 HBs per molecule, explain your results.

d. **Water self-diffusion:** We can evaluate water diffusion by means of the 'RMSD Trajectory Tool'. We learned that:

$D = (1/6) \lim_{t \to \infty} RMSD(t)/t$

For the plot of RMSD(t), evaluate D.

FINAL REMARKS:

1. Play with 'Graphics' -> 'Representations' to customize your view.
2. You can save a snapshot of your system at 'File' -> 'Render'
3. And you can save a movie of your simulation with tool 'Movie Maker' from 'Extensions' -> 'Visualization'.

STEP 4: EXTENSIONS using CHARMM-GUI

1. In order to extend your study to other classes of CNT, such as (10,10) or (8,10), etc. or even consider a planar graphene sheet, you must prepare a new input. The best tool for this is the facility "CHARMM-GUI" (https://www.charmm-gui.org). You must first create a user.
2. Go to 'Input generator' -> 'Nanomaterial Modeler' -> Select 'Nanomaterial Type' = 'Carbonaceous Materials' -> carbon nanotube' or 'graphene'. There you can select the class of tube (for instance (10,10)). Then take "solvated" and "next step".
3. Do not consider "Include ions", only "water", choosing "Fit waterbox size to material size" and "next step" twice.
4. Select "Input generator" -> NAMD and go on.

5. Download “charm-gui.tar”, package including all files you need for the simulations.
6. First run file “step4_equilibration.inp” for equilibration.
7. Second, run file “step5_production.inp” for production.
8. Once finished, you’ll be able to analyze the new trajectory.

QUESTIONS:

1. Based on what you learned up to now, explore the **influence of temperature** on microscopic properties of water. Hint: Try equilibration + “short” production runs (about 300000 steps) at 283 and 323 K.

2. What is the effect of the size on your simulations? Try smaller cases (short CNT and less water). To do this, you should use the tool CHARMM-GUI.

3. How could you make the simulations more realistic? What particles could be included? Try a case (for instance N = 50, T = 303 K) including such particles. To do this, prepare a setup with CHARMM-GUI.